

## BZIP2 vs Unix Compress

| Aspect                       | BZIP2 Format                                                                                       | Unix Compress Format                                     |
|------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Full Form</b>             | BZIP2 (no expansion, just name)                                                                    | Unix compress utility                                    |
| <b>Primary Purpose</b>       | High compression ratio, efficient for large files                                                  | Simple, fast compression, standard in early Unix systems |
| <b>File Extension</b>        | .bz2 (or .tar.bz2 when combined with TAR)                                                          | .Z                                                       |
| <b>Compression Algorithm</b> | Burrows–Wheeler Transform (BWT) + Move-to-Front (MTF) + Run-Length Encoding (RLE) + Huffman coding | LZW (Lempel–Ziv–Welch) dictionary-based compression      |
| <b>Multi-file Support</b>    | No – compresses single file (use TAR for multiple files)                                           | No – compresses single file (use TAR for multiple files) |
| <b>Structure</b>             | Block-based (typically 100–900 KB blocks) with transformations and entropy coding                  | Stream-based LZW encoding with simple header             |
| <b>Error Checking</b>        | CRC-32 per block                                                                                   | No strong error detection (basic checks only)            |
| <b>Metadata</b>              | Stores block size, CRC checks, compression settings                                                | Minimal (just file name and size)                        |
| <b>Encryption Support</b>    | No (requires external tools)                                                                       | No                                                       |
| <b>Compression Ratio</b>     | Very high (better than GZIP for text and structured data)                                          | Moderate, worse than modern methods                      |
| <b>Compression Speed</b>     | Slower compared to GZIP/ZIP                                                                        | Very fast (but less efficient)                           |
| <b>Decompression Speed</b>   | Slower than GZIP (due to complex transforms)                                                       | Very fast                                                |
| <b>Platform Usage</b>        | Linux/Unix systems, archives of source code, scientific data, backups                              | Early Unix systems, mostly obsolete today                |
| <b>Typical Command</b>       | bzip2 file.txt → file.txt.bz2<br>bunzip2 file.txt.bz2                                              | compress file.txt → file.txt.Z<br>uncompress file.txt.Z  |
| <b>Applications</b>          | Used in Linux distributions, software packages, archival storage                                   | Historical Unix file compression, rarely used today      |